

URBAN HEAT ISLAND ANALYSIS

Kitsap County, Washington, United States

Date: July 1, 2026

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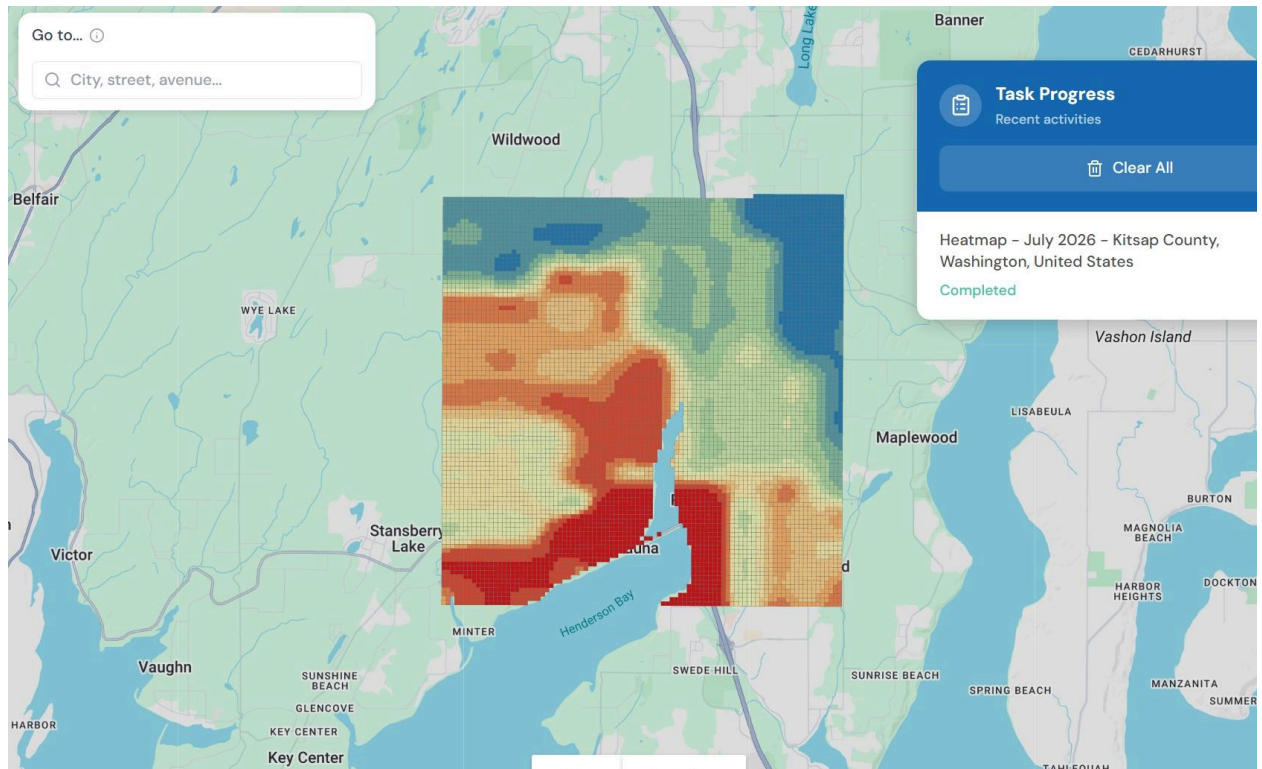
Subject: Geography / Environmental Science

Data Source: FortyGuard Dashboard

1. INTRODUCTION

An Urban Heat Island is an area where a city or town is significantly warmer than its surrounding rural areas. This happens because of concrete, asphalt, buildings, and lack of vegetation. These materials absorb and retain heat during the day and release it at night.

The purpose of this study is to analyze temperature variation in Kitsap County, Washington, USA using satellite heatmap data.



2. METHODOLOGY

Data was collected from the FortyGuard Dashboard for Kitsap County, Washington on July 1, 2026.

- Location: Purdy, Burley, Gig Harbor, WA
- Granularity: 100x100 meters
- Data Type: Surface Temperature Heatmap
- Tool Used: FortyGuard Web Dashboard

Heatmap Accuracy



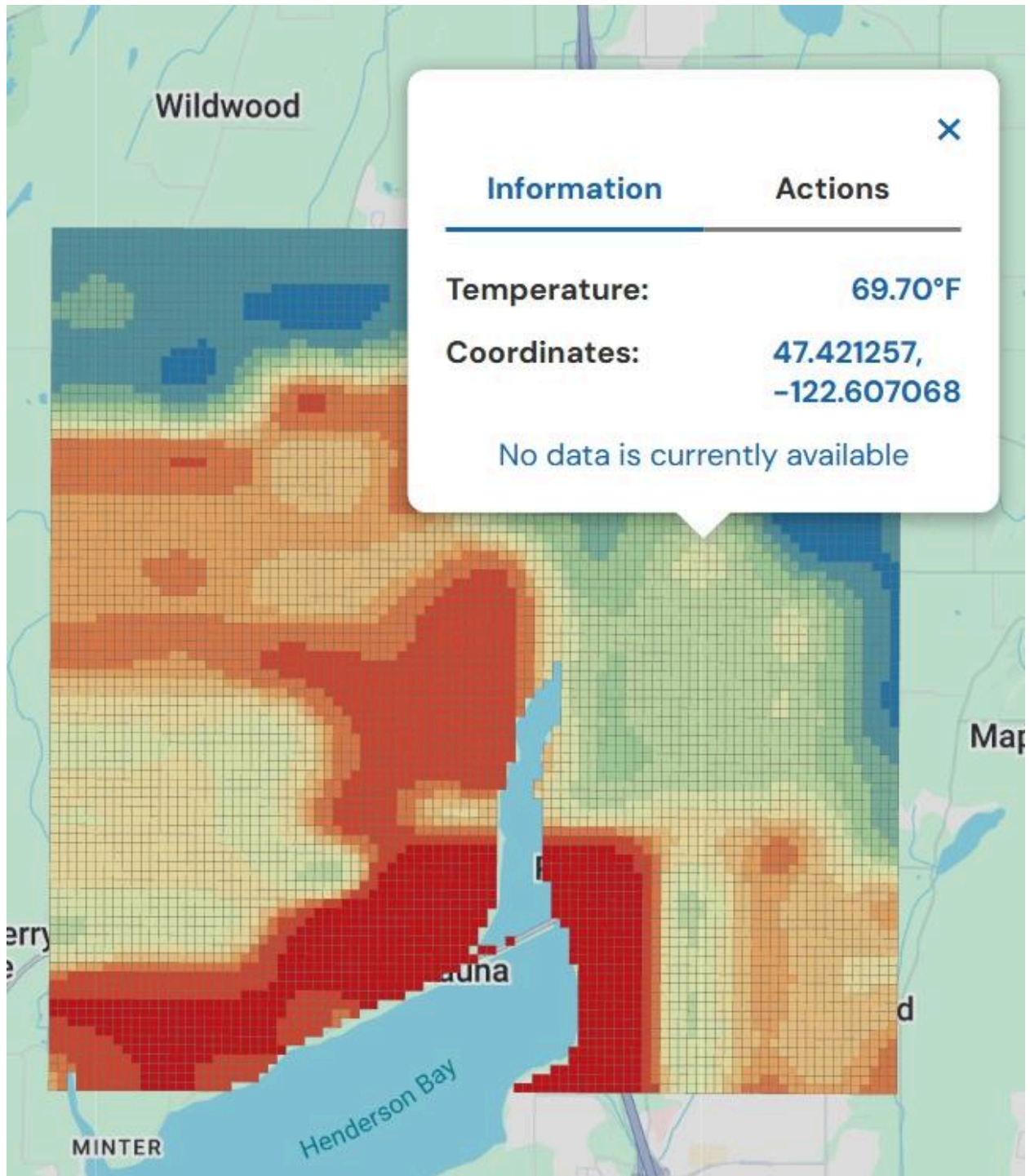
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 **Temperature Range:** 67.49°F – 74.39°F

Tier	Description
1 – Most ideal scenario	High data availability ensures top accuracy for decision-making.
2	Slightly reduced data availability, still very reliable for most purposes.
3	Limited accuracy from older data is useful for broader trends and historical analysis but some nuances may be overlooked.
4	Good data availability provides fairly accurate short-term forecasts.
5	Moderate data leads to reliable, but less precise predictions.
6	Broader trends can be predicted, though fine details are lacking.

3. RESULTS

- Figure 1: Surface Temperature Heatmap - Kitsap County, July 2026*



- Table 1: Temperature Statistics*

Metric Value

Maximum Temperature 74.39°F

Minimum Temperature 67.49°F

Temperature Range 6.9°F

Average Temperature 69-70°F

Heatmap Statistics



Heatmap Statistics



Maximum Temperature

74.39°F

Minimum Temperature

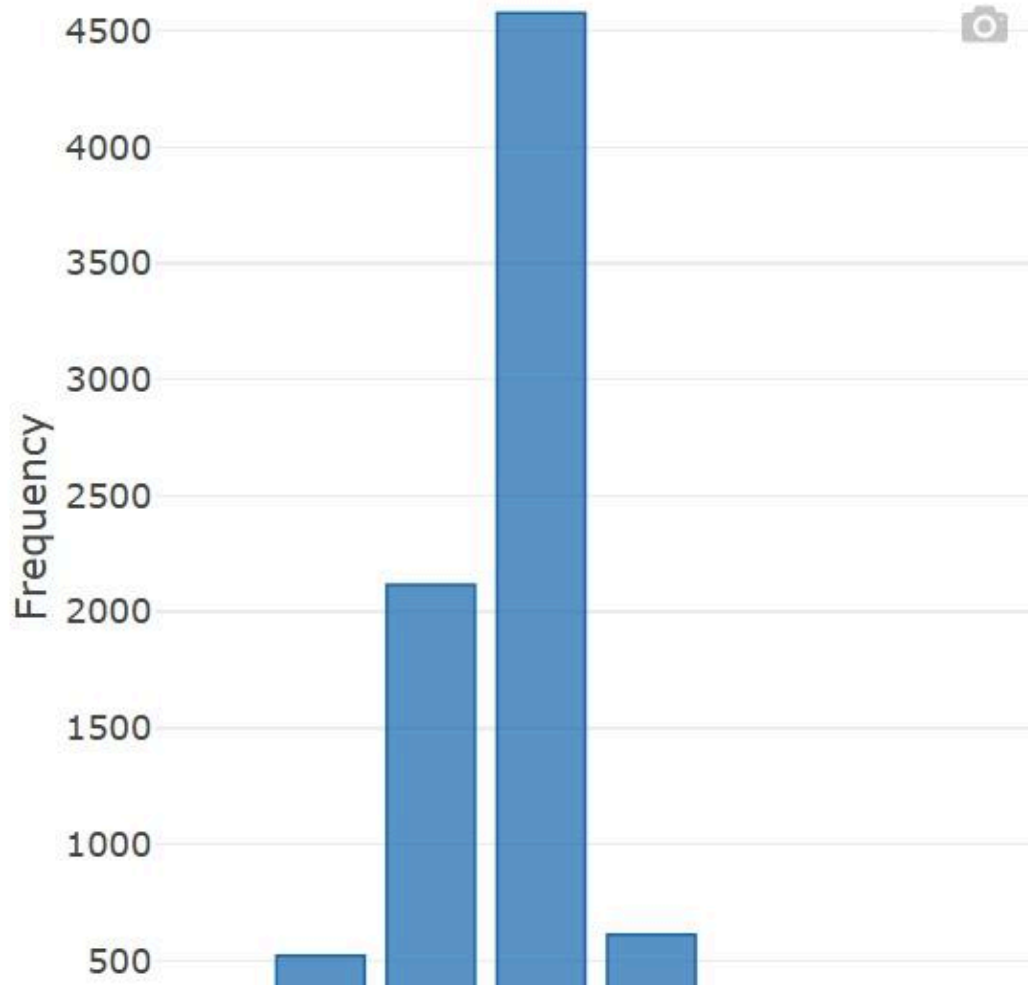
67.49°F

Ter

6.



Bar Graph



4. ANALYSIS

The heatmap shows clear temperature differences across Kitsap County.

1. ***Hot Zones - Red/Orange***: Areas around Purdy, Wauna, and major roads like Highway 16 show the highest temperatures of 74.39°F. This is due to asphalt, concrete, and buildings that trap heat.
2. ***Cool Zones - Blue/Green***: Areas near Henderson Bay and vegetated regions show the lowest temperatures of 67.49°F. Water and trees have a natural cooling effect.
3. ***Conclusion from Data***: There is a 6.9°F temperature difference within the same county. This proves that urban infrastructure directly causes higher local temperatures, known as the Urban Heat Island effect.

Heatmap Metadata



Navigation icons: back, forward, search, etc.

Time Frame

Single Month

Date

Jul 01, 2026

Start Date

Jul 01, 2026

End Date

Jul 31, 2026

Start Time

12:00 AM

End Time

11:00 PM

Spatial Information

Granularity

100m x 100m

Heatmap Area

33.46 mi²

Perimeter

30.84 mi

Temperature Information

● Minimum

57.50

● Maximum

74.40

5. CONCLUSION

This study confirms the presence of Urban Heat Islands in Kitsap County, WA. Built environments are up to 6.9°F hotter than nearby water and green spaces. To reduce this effect, the county can plant more trees, create parks, and use reflective "cool roofs" on buildings.

6. REFERENCES

Data provided by FortyGuard Dashboard, July 2026.